MSG No. 221.95 $Pm\bar{3}m'$ [Type III, cubic]

Table 1: Wyckoff site: `1a`, site symmetry: `m-3m'`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `1b`, site symmetry: `m-3m'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 3: Wyckoff site: `3c`, site symmetry: `4'/mm.m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16,25,26,33,34,37,38,45,46]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,10,12,17,20,22,24,29,30,31,32,41,42,43,44]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,9,11,18,19,21,23,27,28,35,36,39,40,47,48]`

Table 4: Wyckoff site: `3d`, site symmetry: `4'/mm.m'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,13,14,15,16,25,26,33,34,37,38,45,46]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,8,10,12,17,20,22,24,29,30,31,32,41,42,43,44]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[6,7,9,11,18,19,21,23,27,28,35,36,39,40,47,48]`

Table 5: Wyckoff site: `6e`, site symmetry: `4'm.m'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,15,16,25,26,45,46]`
2	$[-x, 0, 0]$	`[3,4,13,14,33,34,37,38]`
3	$[0, x, 0]$	`[5,12,20,22,29,31,42,44]`
4	$[0, 0, x]$	`[6,9,19,23,28,35,39,48]`
5	$[0, 0, -x]$	`[7,11,18,21,27,36,40,47]`
6	$[0, -x, 0]$	`[8,10,17,24,30,32,41,43]`

Table 6: Wyckoff site: `6f`, site symmetry: `4'm.m'`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,15,16,25,26,45,46]`
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,13,14,33,34,37,38]`
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[5,12,20,22,29,31,42,44]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	`[6,9,19,23,28,35,39,48]`
5	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	`[7,11,18,21,27,36,40,47]`
6	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[8,10,17,24,30,32,41,43]`

Table 7: Wyckoff site: `8g`, site symmetry: `.3m'`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6,44,46,48]`
2	$[x, -x, -x]$	`[2,10,11,40,41,45]`
3	$[-x, x, -x]$	`[3,7,12,37,42,47]`
4	$[-x, -x, x]$	`[4,8,9,38,39,43]`
5	$[-x, -x, -x]$	`[13,17,18,32,34,36]`
6	$[-x, x, x]$	`[14,22,23,28,29,33]`
7	$[x, -x, x]$	`[15,19,24,25,30,35]`
8	$[x, x, -x]$	`[16,20,21,26,27,31]`

Table 8: Wyckoff site: `12h`, site symmetry: `2mm..`

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	`[1,2,15,16]`
2	$\left[-x, \frac{1}{2}, 0\right]$	`[3,4,13,14]`
3	$\left[0, x, \frac{1}{2}\right]$	`[5,12,20,22]`
4	$\left[\frac{1}{2}, 0, x\right]$	`[6,9,19,23]`
5	$\left[\frac{1}{2}, 0, -x\right]$	`[7,11,18,21]`
6	$\left[0, -x, \frac{1}{2}\right]$	`[8,10,17,24]`
7	$\left[x, 0, \frac{1}{2}\right]$	`[25,26,45,46]`
8	$\left[0, \frac{1}{2}, -x\right]$	`[27,36,40,47]`
9	$\left[0, \frac{1}{2}, x\right]$	`[28,35,39,48]`
10	$\left[\frac{1}{2}, x, 0\right]$	`[29,31,42,44]`
11	$\left[\frac{1}{2}, -x, 0\right]$	`[30,32,41,43]`
12	$\left[-x, 0, \frac{1}{2}\right]$	`[33,34,37,38]`

Table 9: Wyckoff site: `12i`, site symmetry: `m.2'm'`

No.	position	mapping
1	$[0, y, y]$	`[1,14,33,46]`
2	$[0, -y, -y]$	`[2,13,34,45]`
3	$[0, y, -y]$	`[3,16,26,37]`
4	$[0, -y, y]$	`[4,15,25,38]`
5	$[y, 0, y]$	`[5,24,30,44]`
6	$[y, y, 0]$	`[6,21,27,48]`
7	$[-y, y, 0]$	`[7,23,28,47]`
8	$[-y, 0, y]$	`[8,22,29,43]`

continued ...

Table 9

No.	position	mapping
9	$[-y, -y, 0]$	`[9,18,36,39]`
10	$[y, 0, -y]$	`[10,20,31,41]`
11	$[y, -y, 0]$	`[11,19,35,40]`
12	$[-y, 0, -y]$	`[12,17,32,42]`

Table 10: Wyckoff site: `12j`, site symmetry: `m.2'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, y, y\right]$	`[1,14,33,46]`
2	$\left[\frac{1}{2}, -y, -y\right]$	`[2,13,34,45]`
3	$\left[\frac{1}{2}, y, -y\right]$	`[3,16,26,37]`
4	$\left[\frac{1}{2}, -y, y\right]$	`[4,15,25,38]`
5	$\left[y, \frac{1}{2}, y\right]$	`[5,24,30,44]`
6	$\left[y, y, \frac{1}{2}\right]$	`[6,21,27,48]`
7	$\left[-y, y, \frac{1}{2}\right]$	`[7,23,28,47]`
8	$\left[-y, \frac{1}{2}, y\right]$	`[8,22,29,43]`
9	$\left[-y, -y, \frac{1}{2}\right]$	`[9,18,36,39]`
10	$\left[y, \frac{1}{2}, -y\right]$	`[10,20,31,41]`
11	$\left[y, -y, \frac{1}{2}\right]$	`[11,19,35,40]`
12	$\left[-y, \frac{1}{2}, -y\right]$	`[12,17,32,42]`

Table 11: Wyckoff site: `24k`, site symmetry: `m..`

No.	position	mapping
1	$[0, y, z]$	`[1,14]`
2	$[0, -y, -z]$	`[2,13]`
3	$[0, y, -z]$	`[3,16]`
4	$[0, -y, z]$	`[4,15]`
5	$[z, 0, y]$	`[5,24]`
6	$[y, z, 0]$	`[6,21]`
7	$[-y, z, 0]$	`[7,23]`
8	$[-z, 0, y]$	`[8,22]`
9	$[-y, -z, 0]$	`[9,18]`
10	$[z, 0, -y]$	`[10,20]`
11	$[y, -z, 0]$	`[11,19]`
12	$[-z, 0, -y]$	`[12,17]`
13	$[0, -z, y]$	`[25,38]`
14	$[0, z, -y]$	`[26,37]`
15	$[z, y, 0]$	`[27,48]`
16	$[-z, y, 0]$	`[28,47]`
17	$[-y, 0, z]$	`[29,43]`
18	$[y, 0, z]$	`[30,44]`
19	$[y, 0, -z]$	`[31,41]`
20	$[-y, 0, -z]$	`[32,42]`

continued ...

Table 11

No.	position	mapping
21	$[0,\ z,\ y]$	`[33,46]`
22	$[0,\ -z,\ -y]$	`[34,45]`
23	$[z,\ -y,\ 0]$	`[35,40]`
24	$[-z,\ -y,\ 0]$	`[36,39]`

Table 12: Wyckoff site: `24l`, site symmetry: `m..`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\ -y,\ -z\right]$	`[2,13]`
3	$\left[\frac{1}{2},\ y,\ -z\right]$	`[3,16]`
4	$\left[\frac{1}{2},\ -y,\ z\right]$	`[4,15]`
5	$\left[z,\ \frac{1}{2},\ y\right]$	`[5,24]`
6	$\left[y,\ z,\ \frac{1}{2}\right]$	`[6,21]`
7	$\left[-y,\ z,\ \frac{1}{2}\right]$	`[7,23]`
8	$\left[-z,\ \frac{1}{2},\ y\right]$	`[8,22]`
9	$\left[-y,\ -z,\ \frac{1}{2}\right]$	`[9,18]`
10	$\left[z,\ \frac{1}{2},\ -y\right]$	`[10,20]`
11	$\left[y,\ -z,\ \frac{1}{2}\right]$	`[11,19]`
12	$\left[-z,\ \frac{1}{2},\ -y\right]$	`[12,17]`
13	$\left[\frac{1}{2},\ -z,\ y\right]$	`[25,38]`
14	$\left[\frac{1}{2},\ z,\ -y\right]$	`[26,37]`
15	$\left[z,\ y,\ \frac{1}{2}\right]$	`[27,48]`
16	$\left[-z,\ y,\ \frac{1}{2}\right]$	`[28,47]`
17	$\left[-y,\ \frac{1}{2},\ z\right]$	`[29,43]`
18	$\left[y,\ \frac{1}{2},\ z\right]$	`[30,44]`
19	$\left[y,\ \frac{1}{2},\ -z\right]$	`[31,41]`
20	$\left[-y,\ \frac{1}{2},\ -z\right]$	`[32,42]`
21	$\left[\frac{1}{2},\ z,\ y\right]$	`[33,46]`
22	$\left[\frac{1}{2},\ -z,\ -y\right]$	`[34,45]`
23	$\left[z,\ -y,\ \frac{1}{2}\right]$	`[35,40]`
24	$\left[-z,\ -y,\ \frac{1}{2}\right]$	`[36,39]`

Table 13: Wyckoff site: `24m`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,44]`
2	$[x,\ -x,\ -z]$	`[2,41]`
3	$[-x,\ x,\ -z]$	`[3,42]`
4	$[-x,\ -x,\ z]$	`[4,43]`
5	$[z,\ x,\ x]$	`[5,48]`
6	$[x,\ z,\ x]$	`[6,46]`
7	$[-x,\ z,\ -x]$	`[7,37]`
8	$[-z,\ -x,\ x]$	`[8,39]`

continued ...

Table 13

No.	position	mapping
9	$[-x, -z, x]$	[9,38]
10	$[z, -x, -x]$	[10,40]
11	$[x, -z, -x]$	[11,45]
12	$[-z, x, -x]$	[12,47]
13	$[-x, -x, -z]$	[13,32]
14	$[-x, x, z]$	[14,29]
15	$[x, -x, z]$	[15,30]
16	$[x, x, -z]$	[16,31]
17	$[-z, -x, -x]$	[17,36]
18	$[-x, -z, -x]$	[18,34]
19	$[x, -z, x]$	[19,25]
20	$[z, x, -x]$	[20,27]
21	$[x, z, -x]$	[21,26]
22	$[-z, x, x]$	[22,28]
23	$[-x, z, x]$	[23,33]
24	$[z, -x, x]$	[24,35]

Table 14: Wyckoff site: 48n, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x, -y, -z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[-x, -y, z]$	[4]
5	$[z, x, y]$	[5]
6	$[y, z, x]$	[6]
7	$[-y, z, -x]$	[7]
8	$[-z, -x, y]$	[8]
9	$[-y, -z, x]$	[9]
10	$[z, -x, -y]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-z, x, -y]$	[12]
13	$[-x, -y, -z]$	[13]
14	$[-x, y, z]$	[14]
15	$[x, -y, z]$	[15]
16	$[x, y, -z]$	[16]
17	$[-z, -x, -y]$	[17]
18	$[-y, -z, -x]$	[18]
19	$[y, -z, x]$	[19]
20	$[z, x, -y]$	[20]
21	$[y, z, -x]$	[21]
22	$[-z, x, y]$	[22]
23	$[-y, z, x]$	[23]
24	$[z, -x, y]$	[24]
25	$[x, -z, y]$	[25]

continued ...

Table 14

No.	position	mapping
26	$[x, z, -y]$	[26]
27	$[z, y, -x]$	[27]
28	$[-z, y, x]$	[28]
29	$[-y, x, z]$	[29]
30	$[y, -x, z]$	[30]
31	$[y, x, -z]$	[31]
32	$[-y, -x, -z]$	[32]
33	$[-x, z, y]$	[33]
34	$[-x, -z, -y]$	[34]
35	$[z, -y, x]$	[35]
36	$[-z, -y, -x]$	[36]
37	$[-x, z, -y]$	[37]
38	$[-x, -z, y]$	[38]
39	$[-z, -y, x]$	[39]
40	$[z, -y, -x]$	[40]
41	$[y, -x, -z]$	[41]
42	$[-y, x, -z]$	[42]
43	$[-y, -x, z]$	[43]
44	$[y, x, z]$	[44]
45	$[x, -z, -y]$	[45]
46	$[x, z, y]$	[46]
47	$[-z, y, -x]$	[47]
48	$[z, y, x]$	[48]